

# JĘDRZEJ WYDRA | RÉSUMÉ

» There are exactly two options: either I find a solution, or I design a new one. «



- » **Status:** PhD student, researcher, and lecturer at the Laboratory of Criminalistics.
- » **Skills:** Solid statistical and ML background with focus on methodology, NLP and prediction.
- » **Fields:** Forensic Science, Machine Learning, Functional Data Analysis, Formal Logic.
- » **Tools:** Python, R, scikit-learn, tidyverse, XGBoost, LaTeX, bash, MS Office, Git, SQL.
- » **Didactics:** Praised by students for making difficult ideas understandable and engaging.
- » **Hobby:** Passionate about psychology, calisthenics, and history of tea.

## Co-Editor and Co-Author – EU AI Act Commentary

Autumn 2025

- » Co-authored and co-edited a comprehensive legal and technical commentary on the EU AI Act.
- » Worked with a multidisciplinary team to analyze implications for AI governance, risk, and transparency.

## Interval Estimation with EM for Forensic Insect Development

June 2025

- » Designed EM-based algorithm for Weibull mixture models with validated interval estimation.
- » More details at [https://github.com/Jedrzej-Wydra/survival\\_analysis\\_em](https://github.com/Jedrzej-Wydra/survival_analysis_em)

## Kernel-based Independence Testing for Functional Data

March 2025

- » Developed high-dimensional test with basis expansion and permutation inference in Python and R.
- » More details at <https://github.com/Jedrzej-Wydra/independence>

## Local Linear Approximation of Nonlinear Functions

February 2025

- » Used gradient descent in sliding-window linear models to simulate local approximation of nonlinear curves.
- » More details at <https://github.com/Jedrzej-Wydra/nonlinear-problem-linear-solution>

## Functional Temperature Reconstruction for Forensic Time Series

December 2024

- » Designed algorithm for reconstructing temperature at crime scene using Fourier and concurrent models.
- » More details at <https://github.com/Jedrzej-Wydra/temperature-reconstruction>

EXPERIENCE  
SELECTED PROJECTS

Explore publications, projects, full résumé, and student reviews at <https://jedrzej-wydra.github.io>

## PhD in law (in progress) (Adam Mickiewicz University)

2020 - current

- » PhD Thesis: *Evaluation of Evidence from Insect-Based Time-of-Death Estimation Using Machine Learning*
- » Combined forensic science, legal reasoning, and machine learning to improve the value of evidence.

## M.Sc. in Theoretical Mathematics (Adam Mickiewicz University)

2020 - 2023

- » Master Thesis: *Good, Evil and Obligation: a multimodal Deontic Logic*
- » Formalized ethical and legal dilemmas using multimodal deontic logic and algebraic structures.

## B.Sc. in Mathematics (spec. Data Analysis) (Adam Mickiewicz University)

2016 - 2020

- » Bachelor Thesis: *Semantics for Deontic Logics*
- » Formalized normative reasoning using deontic logic with both relational and algebraic semantics.

## Master's Degree in Law (spec. Forensic Science) (Adam Mickiewicz University)

2015 - 2020

- » Master Thesis: *Evaluation of Evidence from Individualizing Analysis of Handwriting*
- » Developed a probabilistic method to assess handwriting evidence with uncertainty quantification.

EDUCATION

**Selected completed courses:** Financial law, Financial arithmetic and portfolio analysis, Economic insurance law, Actuarial mathematics